

MAHESWARA REDDY

CONTACT

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OBJECTIVE

Telecommunications Software Engineer with 5+ years of hands-on experience in SIP/VoIP technologies, FreeSWITCH, C/C++, and Golang, with expertise in call-control architecture, event-driven routing, voice application integration, and performance optimization. Looking to contribute to challenging engineering roles focused on designing and developing scalable communication platforms, solving complex real-time system problems, and driving technical excellence through end-to-end ownership.

EXPERIENCE

Insemi Technology Pvt. Ltd

May 2026 - Present

Staff Engineer

Dec 2024 - April 2026

Staff Software Engineer
- Contributed to the design and development of SIP/VoIP-based communication and IVR solutions using FreeSWITCH, supporting scalable call-center and real-time communication use cases.
- Developed and implemented call-routing and event-driven logic using FreeSWITCH ESL and Golang, including the routing component developed independently to support dynamic call-flow requirements.
- Worked on SIP signaling, FreeSWITCH dialplans, ESL, IVR call flows, RTP/media handling, and MRCP integrations, contributing to POCs and solution development for client requirements.
- Collaborated with clients and cross-functional engineering teams to understand technical requirements, troubleshoot complex call-flow and integration issues, and deliver functional - POCs demonstrating solution feasibility.
Independently designed and implemented - Ansible automation and deployment playbooks for containerized applications, integrating deployment workflows with Docker and Rio CI/CD pipelines.
- Contributed to performance, scalability, troubleshooting, and deployment activities for communication platforms, including testing and optimization of high-concurrency FreeSWITCH-based solutions.

BRRaySoft Solutions Pvt Ltd

May 2024 - Dec 2024

Senior Consultant
- Worked as a Senior Consultant at infosys for VoIP SIP FreeSWITCH Application development.
- SIP-based Freeswitch solutions, driving innovation and experimentation
- Worked closely with cross-functional teams to design and deliver POCs, showcasing technical expertise and creativity
- Contributed to the development of best practices and standards for POC development within the organization.

Mirafra Software Technologies, Hyderabad

4th Dec 2023 - May 2024

Software Engineer II
Worked as a Software Engineer II at Mirafra Technologies.

SKILLS

Programming Languages : C, C++, Shell scripting, Python, Golang.

Platforms/OS : Linux

Protocols & Networking: SIP Signaling, Socket Programming, HTTP, WebSockets, SDP, RTP, TCP, UDP, WSS, SCTP.

Domain & Concepts : VoIP SIP, Telecom/Networking, Multithreading Data Structures, PostgreSQL.

Defect Tracking : GIT, JIRA,Radar.

Debugging Tools &IDEs : Valgrind, GDB, TcpDump, Wireshark, Visual Studio Code ,SIPP.

Build System : Make, CMake

Experience in developing Media servers that handles decoding, encoding, and transcoding of H.264, VP8, VP9 Codecs.

Experience using protocol RTP/RTCP and implementation of REMB (NACK/PLI/FIR)

Relevant experience in developing Media server (SFU/MCU) using Gstreamer and WebRTC for Video streaming at server-side stream routing, transcoding, or Video mixing.

DevOps & Automation: Docker, Ansible Automation Platform, RIO based CI/CD Pipelines, Infrastructure Automation

LANGUAGES

Telugu - Read/Write/Speak

English - Read/Write/Speak

Vectone Mobile India Services Pvt Ltd, Chennai

4th Jan 2021 - 16th Oct 23

Software Engineer - C++
Experienced Software Engineer adept in C++ development, bringing 2.10 years of expertise in delivering efficient and innovative solutions in telecom or media Domain.

Vector India Embedded Training Institute, Chennai

Aug 2019 - Jan 2020

Embedded Trainee
Experienced Embedded softwares and learnt programming concepts like C,C++, Embedded C and communication protocols like I2C, SPI, UART, CAN. and certified a Embedded Trainee.

PROJECTS

Next-Gen Call Center Platform using FreeSWITCH

- Design and development a scalable SIP-based Freeswitch solution for a next-gen call center platform.
- Integrated with multiple third-party APIs for enhanced functionality.
- Collaborated with cross-functional teams to deliver a working POC, demonstrating the solution's feasibility and potential.

Technologies used:- FreeSWITCH, SIP, TCP/IP, C, C++, Python, Golang, Shell Scripting, Ansible Automation platform for deployments.
Platform:- Linux

Tile arrangement in video processing using Gstreamer

This project involves arranging multiple video streams into a grid or customized layout.
Key features:- Video Input Handling,Dynamic Tile Layout logic implementation,Error Handling and Stability.
Challenges and considerations :-
Managing multiple video streams concurrently without compromising performance.
Synchronizing the start, pause, and stop functionalities across various video inputs.

Technologies used:- Gstreamer, UDP, C, C++.
Platform:- Linux

DTLS Reconnection B/W Client and server

This project was actually try to reconnect the client and media server through DTLS, handshake on first connection failure, when it fails, based on the notification sent via socket messages and client will send and SIP reinvite to the Server so that Media server gets connected and will successfully send and receive the packets after negotiation.

Technologies used:- C++, SIP, DTLS, RTP, Gstreamer
Platform:- Linux

Encode and Decode JSON Message

This project describes make communication between server over TCP, Encode the JSON Message from relevant structure and decide JSON message to relavant Structure by parsing the variables.

Technologies used: C, C++, TCP, JSON
Platform:- Linux

Video Meet - Audio and Video Recording

This Project Describes Record the audio and video streams and mix them and store in NFS for security reasons and make them available for the users when they needed, it will be used for students and company private meetings etc., Suppose there are some cases when someone not able to attend the meeting and record the video allowed by host, those will be available to all the users even after certain time period based on the plan that we have provided to the users.

Technologies used:- C++, Audio/Video Encoding/Decoding, NFS, GStreamer, WebRTC
Platform:- Linux

COUNT INCOMING/OUTGOING BUFFER IN PIPELINE USING GSTREAMER.

This project describes the buffer which we have in gstreamer pipeline passes through each and every element could able to count using the probe concept in gstreamer. Here we will get the Buffer count and also we can manipulate the RTP packet header by extraction of buffer.

Technologies used:- C, C++, Gstreamer,RTP
Platform:- Linux

MP4 FILE DURATION FINDER FOR VIDEO RECORD USING GSTREAMER

Using gstreamer and FFMpeg we can find the duration, size of File.

Technologies used:- C, C++, GStreamer, FFMpeg
Platform:- Linux

SIP over SCTP

This project describes the usage of delivering the large size messages reliably and to reduce the chance of out of order delivery and message loss as well. So with this we have overcome the message loss when sending large size of sdp information in SIP over SCTP, compared to TCP, SCTP was stably working without any message loss when we try to send more between nodes, so at the end the data has been successfully sent over SCTP protocol.

Technologies used:- SIP, SCTP, TCP/IP, C, C++.
Platform:- Linux

Weather monitoring system

Developed an embedded monitoring system using ARM Cortex-M controlller and Implemented the sensor interfacing(I2C,UART) for temporature,humidity,and pressure sensors, utilised embedded C to handle the processing and display the reading on LCD.

ACADEMIC CAREER

B.Tech From KSRM College of Engineering ,Kadapa with First class passed out in 2019

Intermediate MPC passed out in 2015 with 81.4% first class in The Nandyal Junior College, Nandyal.

ADDITIONAL INFORMATION

Date:
Place:

CERTIFICATIONS

Professional Embedded Training in Vector Institute (Aug 2019 to Jan 2020).

Won 3rd Prize in Presentation on Digital India at NBKR College of Engineering in Nellore.

DECLARATION

I hereby declare that all the facts given above are true and correct to the best of my knowledge.



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